

Bacterial Cellulose Biosynthesis: The c-di-GMP-Gated Bcs Synthase System, with a Focus on the *Pseudomonas putida* KT2440 Module

A commissioned review-style synthesis (UniPathway UPA00694: BcsA / BcsB / BcsC-like PP_2638)

1. Executive Summary

Bacterial cellulose biosynthesis is a **synthase-dependent, cyclic-di-GMP-gated exopolysaccharide (EPS) secretion pathway** in which a *single* processive glycosyltransferase — the GT-2 family enzyme **BcsA** — performs two coupled tasks: it polymerizes UDP-glucose into β -1,4-glucan and simultaneously translocates the nascent chain across the inner membrane through a channel formed by its own transmembrane helices. This economy of function is the defining feature of the system and distinguishes it from Wzx/Wzy- or ABC-transporter-dependent EPS pathways, in which polymerization and export are handled by separate machines. BcsA is catalytically inert until the second messenger **c-di-GMP** binds its C-terminal PilZ domain, breaking a salt bridge that tethers an autoinhibitory "gating loop"; this allosteric release is the master on/off switch that ties cellulose production to the broader motile-to-sessile lifestyle transition.

Around this catalytic core, a small set of accessory proteins builds a **trans-envelope conduit**. **BcsB**, a membrane-anchored periplasmic subunit, is obligately associated with BcsA and helps organize the periplasmic passage of the glucan chain; in Gram-negative systems a **BcsC-like** protein (a 16-stranded outer-membrane β -barrel preceded by a long periplasmic tetratricopeptide-repeat (TPR) domain — annotated as **PP_2638** in *P. putida* KT2440) completes secretion across the outer membrane. The order of events is physically constrained and unidirectional: **polymerization → inner-membrane translocation → periplasmic handoff → outer-membrane export**.

For the specific working scope — *Pseudomonas putida* KT2440 — the module is a "**Wss/Pseudomonas-type**" **bcs operon** that additionally carries an O-acetylation cassette (WssFGHI), so the secreted polymer is **partially acetylated cellulose**. Its transcription is repressed by the master regulator FleQ (with FleN) and de-repressed by c-di-GMP, embedding the pathway in the same signalling logic that governs flagellar motility. Physiologically, cellulose in *P. putida* plays a comparatively minor role in biofilm bulk but contributes measurably to rhizosphere colonization and survival under water-limiting (desiccation) stress. The mechanistic backbone (BcsA–BcsB catalysis, c-di-GMP activation, BcsC export) is strongly supported by crystallography, cryo-EM, in-crystallo enzymology, and reconstitution; the *P. putida*-specific assignments (notably the precise function of PP_2638 and the adaptive rationale for acetylation) rest largely on homology and remain to be tested directly.

2. Definition and Biological Boundaries

2.1 What is included

The system under review is the **Bcs (bacterial cellulose synthase) synthase-dependent secretion pathway**, comprising three functional steps mapped to the UniPathway UPA00694 module:

1. **Polymerization** of β -1,4-glucan from UDP-glucose (BcsA, GT-2 catalytic subunit).
2. **c-di-GMP-responsive regulation / co-translocation** across the inner membrane (BcsA PilZ domain and BcsB).
3. **Outer-membrane export** of the cellulosic chain (BcsC-like porin/TPR protein; PP_2638 in *P. putida*).

In *Pseudomonas putida* KT2440 the module belongs to the **Wss/Pseudomonas-type** operon, which appends an **O-acetylation** activity (WssFGHI) that chemically modifies the polymer.

2.2 What should be treated separately (neighboring / commonly confused processes)

- **Plant cellulose synthesis (CESA rosettes)**. Plant CESA enzymes share the GT-2 catalytic fold and glucan-forming chemistry with BcsA, but they operate as large membrane "rosette" complexes guided by cortical microtubules, use sucrose-derived UDP-glucose, and are not c-

di-GMP-gated. Crystallographic/modeling work explicitly cautions that plant CESAs are similar in catalytic core but differ in assembly and regulation (PMID: 24139443).

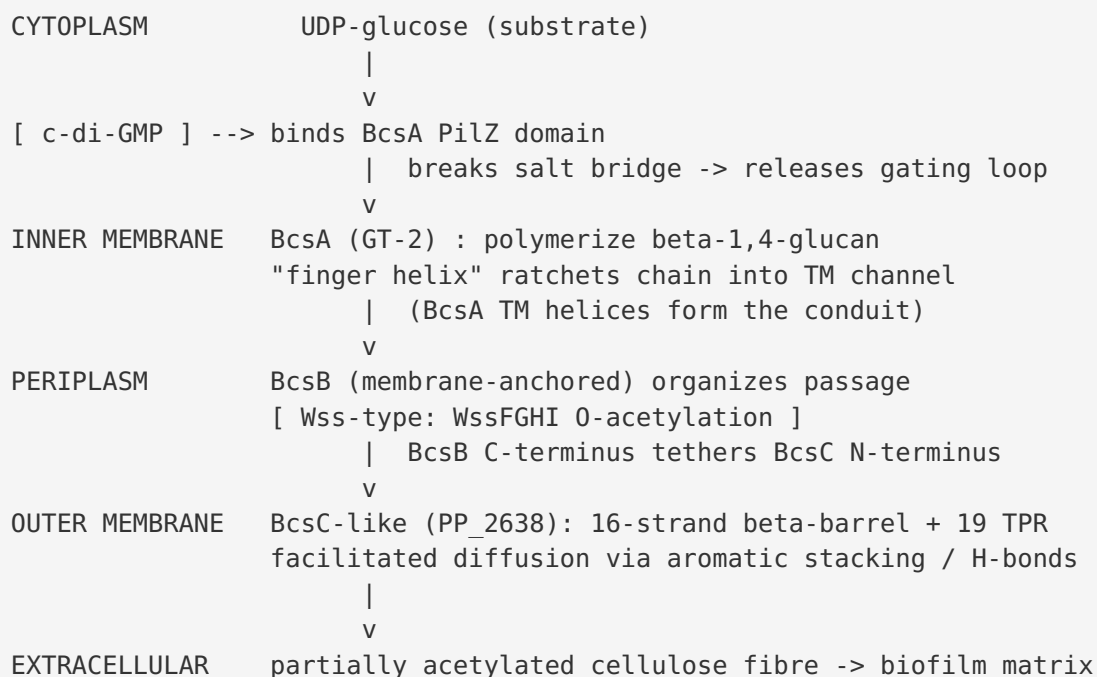
- **Mixed-linkage (1,3;1,4)- β -D-glucans.** These are made by distinct Cellulose-Synthase-Like (CSL) enzymes in grasses and by some bacteria (e.g., *Romboutsia ilealis*), and are chemically and enzymatically separate from pure β -1,4 cellulose (PMID: 37500617).
- **Chemically modified cellulose variants.** *Phosphoethanolamine (pEtN) cellulose* produced by some Enterobacteriaceae uses the same synthase core but a different modification/secretion module (PMID: 38645035; PMID: 39242554). *O-acetylated cellulose* (Wss-type) is the relevant variant for *P. putida*.
- **Alginate biosynthesis (Alg system).** Alginate is another synthase-dependent EPS in *Pseudomonas*, and its acetylation machinery (AlgFIJ) is the evolutionary template for the cellulose acetylation cassette (WssGHI); the two pathways are frequently discussed together but produce chemically distinct polymers (PMID: 28258142).
- **c-di-GMP signalling per se.** c-di-GMP is a global second messenger regulating dozens of biofilm and motility outputs; the cellulose pathway is one *effector* of this network, not the network itself.

2.3 Competing definitions

The main definitional tension is **nomenclature and operon architecture**. The catalytic subunits were historically named BcsA/BcsB and later Cesa/CesB in the acetic-acid-bacterial literature (PMID: 25285473). Operon composition also varies: the *Gluconacetobacter/Komagataeibacter* "Type I" operon includes a **BcsD** crystallinity factor and multiple bcs operons of which only one is complete (PMID: 29674724); enterobacterial "Type II" systems add BcsE/F/G/R/Q and pEtN modification; and the **Wss/Pseudomonas-type** system adds acetylation subunits. Thus "bacterial cellulose synthase" is best defined by the **conserved BcsA–BcsB catalytic/translocation core**, with modification and export modules treated as lineage-specific elaborations.

3. Mechanistic Overview

The best-supported model is a **strictly ordered, four-stage, single-pass secretion process**:



Obligatory steps: (i) UDP-glucose polymerization by BcsA; (ii) c-di-GMP binding as the activating trigger; (iii) inner-membrane translocation coupled to elongation; (iv) outer-membrane export via the BcsC porin in Gram-negatives. Elongation and translocation are **mechanistically coupled** — the polymer cannot be made without being simultaneously threaded through the channel, because the growing chain occupies the translocation pathway.

Conditional/accessory steps: O-acetylation (WssFGHI) is *conditional* — it modifies but is not required to make the polymer; non-acetylated cellulose is still synthesized when the cassette is disrupted (PMID: 14507360). Crystallinity control (BcsD, cytoskeletal scaffolds) is *accessory* and lineage-specific (PMID: 38688160). Transcriptional derepression by c-di-GMP/FleQ is an upstream *gating* layer conditional on physiological state.

Catalytic detail. Each glucose addition is modeled as an **SN2-type single-displacement** reaction; QM/MM calculations estimate an activation barrier of ~68 kJ/mol and a per-addition rate of ~8 s⁻¹ (PMID: 25942604). In-crystallo enzymology directly visualized a **full biosynthetic cycle** and defined a **"finger helix" ratcheting mechanism** that pushes the terminal glucose into the transmembrane channel after each addition (PMID: 26958837).

4. Major Molecular Players and Active Assemblies

4.1 BcsA — the catalytic/translocation engine (Step 1)

Finding F001 & F002. BcsA is a membrane-integrated GT-2 glycosyltransferase that constitutes the minimal catalytic core together with BcsB. The landmark crystal structure of the *Rhodobacter sphaeroides* BcsA–BcsB complex captured a **translocating polysaccharide** inside the enzyme, establishing that BcsA both polymerizes UDP-glucose into β -1,4-glucan and forms the transmembrane conduit for the nascent chain ([PMID: 23222542](#)):

"Cellulose synthesis and transport across the inner bacterial membrane is mediated by a complex of the membrane-integrated catalytic BcsA subunit and the membrane-anchored, periplasmic BcsB protein."

The **c-di-GMP switch** operates on BcsA's C-terminal PilZ domain. Structures of the c-di-GMP-bound complex show that the ligand **breaks a salt bridge that tethers a conserved gating loop**, thereby relieving autoinhibition and granting substrate access to the active site; engineered disruption of the salt bridge produces a **constitutively active** synthase — a clean genetic confirmation of the allosteric model ([PMID: 24704788](#)):

"c-di-GMP releases an autoinhibited state of the enzyme by breaking a salt bridge that otherwise tethers a conserved gating loop that controls access to and substrate coordination at the active site."

Functional reconstitution independently supports the requirement for c-di-GMP: co-expression of *Gluconacetobacter* Csa/CesB (BcsA/BcsB) in *E. coli* yielded cellulose-synthesizing activity **only when a diguanylate cyclase was also expressed** to supply c-di-GMP ([PMID: 25285473](#)).

4.2 BcsB — the periplasmic regulatory/organizing subunit (Step 2)

BcsB is a membrane-anchored, largely periplasmic partner that is obligately associated with BcsA in the catalytic core ([PMID: 23222542](#)). Its role extends beyond a passive anchor: recent cryo-EM of the *E. coli* system shows that **multiple BcsB copies assemble into a periplasmic semicircle**, whose terminal subunit **tethers the N-terminus of a single BcsC**, coupling inner-membrane catalysis to outer-membrane export ([PMID: 39242554](#)):

"the semicircle's terminal BcsB subunit tethers the N-terminus of a single BcsC protein in a trans-envelope secretion system."

In the working scope, **PSEPK bcsB** is annotated as the c-di-GMP-responsive regulatory subunit with cyclic-di-GMP binding activity. It is worth noting the field's mechanistic center of gravity places the *primary* c-di-GMP sensor at BcsA's PilZ domain; the precise, independent contribution of BcsB's own c-di-GMP responsiveness in *P. putida* is inferred from homology and not yet directly resolved.

4.3 BcsC-like PP_2638 — the outer-membrane exporter (Step 3)

Finding F003. In Gram-negative systems, secretion across the outer membrane is completed by a **BcsC-like protein**. The crystal structure of a truncated BcsC reveals a **16-stranded β -barrel** outer-membrane channel with a gating loop, preceded by a long periplasmic domain built from **19 tetratricopeptide repeats (TPR)** ([PMID: 31604608](#)):

"Translocation across the outer membrane occurs through the BcsC porin, which extends into the periplasm via 19 tetra-tricopeptide repeats (TPR)."

The channel lumen is lined with hydrophilic and aromatic residues, suggesting **facilitated diffusion** of the glucan via aromatic stacking and hydrogen bonding rather than active transport. The cryo-EM trans-envelope structure ties this porin to the BcsB semicircle, forming a continuous secretion conduit ([PMID: 39242554](#)). In *P. putida* KT2440, **PP_2638** is the operon C (BcsC-like) protein fulfilling this export/support role; its assignment is by orthology, and its detailed structure/function has not been characterized directly in this organism.

4.4 The Wss acetylation cassette — a *Pseudomonas*-specific elaboration

Finding F005. *P. putida* KT2440 carries the **Wss/Pseudomonas-type** operon, which adds O-acetylation to the polymer. In *P. fluorescens* SBW25, polar and non-polar mutations in **wssF** or the downstream **wssGHIJ** genes abolish acetylation and yield **non-acetylated cellulose**, genetically implicating this cassette in modification ([PMID: 14507360](#)):

"Both polar and non-polar mutations in the sixth gene of the wss operon (wssF) or adjacent downstream genes (wssGHIJ) generated mutants that overproduce non-acetylated cellulose, thus implicating WssFGHIJ in acetylation of cellulose."

Biochemically, purified **WssI** (from *P. fluorescens* and *Achromobacter insuavis*) is an **O-acetyl(ester)transferase / acetylerase** that acts on cello-oligomers with multiple acetyl-donor substrates (PMID: 37224964). WssGHI are homologues of the alginate acetylation machinery **AlgFIJ** (PMID: 28258142), indicating a shared modular origin. That *P. putida* KT2440 carries this acetylation-competent operon is documented in comparative operon analyses (PMID: 28131895).

4.5 Summary table of players

Step	Protein (PSEPK)	Family / fold	Molecular role	Evidence strength
1. Polymerization + IM translocation	bcsA	GT-2 glycosyltransferase, membrane-integral	Polymerizes UDP-Glc; forms TM channel; PilZ c-di-GMP sensor	Strong (crystallography, in-crystallo enzymology, QM/MM, reconstitution)
2. c-di-GMP regulation / periplasmic organization	bcsB	Membrane-anchored periplasmic	Obligate BcsA partner; organizes periplasmic passage; couples to BcsC	Strong for core role; <i>P. putida</i> -specific c-di-GMP role inferred
3. OM export	PP_2638 (BcsC-like)	16-strand β -barrel + 19 TPR	Facilitated OM export of glucan	Strong for BcsC family; PP_2638 assignment by homology
Modification (conditional)	wssFGHI	AlgFIJ-like acetyltransferases	O-acetylation of cellulose	Strong genetically/ biochemically in <i>P. fluorescens</i>

5. Evolutionary and Cell-Biological Variation

5.1 The ancient, conserved core

The **GT-2 catalytic/translocation module of BcsA** is **deeply conserved**, spanning bacteria to plant CESA enzymes. Structural and modeling analyses show that the cotton CESA catalytic domain can be modeled on the bacterial BcsA scaffold, underscoring a shared ancestral chemistry for β -1,4-glucan synthesis coupled to membrane translocation ([PMID: 24139443](#)). BcsA is therefore the **best representative for understanding the ancestral role** of the cellulose-synthase family; the enterobacterial pEtN-cellulose and plant rosette systems are best read as lineage-specific elaborations built on this core.

5.2 Operon-level variation across lineages

- **Acetic acid bacteria (*Gluconacetobacter/Komagataeibacter*, Type I).** Multiple bcs operons, of which typically only one is complete (bcsABCD); presence of **BcsD** and cytoskeletal guidance yields highly crystalline cellulose I; these are the premier industrial producers ([PMID: 29674724](#); [PMID: 29926141](#)).
- **Enterobacteria (Type II; *Salmonella*, *E. coli*).** Add BcsEFG-RQ and produce **pEtN-modified** cellulose that, with curli, forms the CsgD-regulated biofilm matrix ([PMID: 38645035](#); [PMID: 41656623](#)).
- **Pseudomonads and select β -proteobacteria (Wss/*Pseudomonas*-type).** Add the **WssFGHI acetylation** cassette. Strikingly, the unrelated respiratory pathogen *Bordetella avium* 197N carries a nearly identical acetylation-competent operon, and GC-content analysis indicates the synthase + acetylation subunits were **acquired as a single horizontally-transferred unit** ([PMID: 28131895](#)).
- **Gram-positive route.** *Romboutsia ilealis* makes mixed-linkage β -glucan via a distinct synthase — an **alternative molecular route** to a β -glucan EPS outcome ([PMID: 37500617](#)).

5.3 Physiological-state variation in *P. putida*

Finding F004. In *P. putida*, the bcs operon is embedded in the c-di-GMP/FleQ regulatory switch. The **bcsD promoter (PbcsD)** is **negatively regulated by FleQ and FleN**, and this repression is **antagonized by c-di-GMP** ([PMID: 30889223](#)):

"PbcsD was negatively regulated by FleQ and FleN, and repression was antagonized by c-di-GMP."

This places cellulose gene expression under the **same master switch that governs the motility-to-sessility transition**: rising c-di-GMP simultaneously derepresses cellulose transcription and allosterically activates BcsA catalysis — a coherent feed-forward that commits cells to a sessile program. Functionally, in *P. putida* mt-2, Bcs plays a **minor role in biofilm bulk/stability** but **contributes to rhizosphere colonization** (competition assays) and is **induced by water stress**, implicating cellulose in desiccation tolerance ([PMID: 21507177](#)):

"Bcs plays a minor role in biofilm formation and stability, although it does contribute to rhizosphere colonization based on a competition assay."

6. Constraints, Dependencies, and Failure Modes

Ordering constraints (hard). The four stages are physically sequential and cannot be reordered:

1. **c-di-GMP must bind before catalysis.** Without ligand, the gating loop occludes the active site; reconstitution fails absent a diguanylate cyclase ([PMID: 25285473](#); [PMID: 24704788](#)).
2. **Polymerization and IM translocation are obligately coupled.** The growing chain occupies the very channel that would transport it; the finger-helix ratchet advances the polymer one glucose at a time ([PMID: 26958837](#)). This rules out any model in which a pre-formed cytoplasmic glucan is exported after synthesis.
3. **Periplasmic handoff precedes OM export.** BcsB must tether BcsC to complete the conduit ([PMID: 39242554](#)).
4. **OM export is facilitated diffusion**, dependent on the aromatic-lined BcsC barrel ([PMID: 31604608](#)).

Conditional / mutually exclusive events. - **Acetylation vs. non-acetylation.** Loss of WssFGHI shifts the product to non-acetylated cellulose without abolishing synthesis, so acetylation is a modification branch, not a gate (PMID: 14507360). - **Motility vs. sessility.** High c-di-GMP favors cellulose/biofilm and suppresses motility; the FleQ node makes these outcomes largely reciprocal (PMID: 30889223).

Failure modes / functional consequences. - **Non-acetylated cellulose weakens biofilm.** In *P. fluorescens* SBW25, air-liquid interface (wrinkly spreader) biofilms **require the acetylated form of cellulose**; biofilm strength further depends on **LPS-cellulose fibre interactions** (PMID: 14507360; PMID: 16151196):

"a major component of the WS air-liquid biofilm strength results from the interactions between LPS and the cellulose matrix of the biofilm."

- **Loss of c-di-GMP signalling** abolishes both transcriptional derepression and catalytic activation — a double failure that silences the pathway.
- **In *P. putida* specifically**, cellulose loss produces only a modest biofilm phenotype but a measurable colonization/desiccation-fitness cost (PMID: 21507177).

7. Controversies and Open Questions

1. **What does BcsB actually do in *P. putida*?** The working annotation assigns c-di-GMP binding to PSEPK bcsB, yet the strongest structural evidence localizes the primary c-di-GMP sensor to BcsA's PilZ domain. Whether *P. putida* BcsB independently senses c-di-GMP, or primarily serves as the periplasmic organizer/BcsC-tether shown in *E. coli*, is unresolved and rests on cross-organism inference.
2. **Function of PP_2638.** The assignment of PP_2638 as the BcsC-like exporter is by homology to structurally characterized BcsC from other Gram-negatives. Direct evidence — knockout phenotype, structure, or export assay in KT2440 — is lacking.
3. **Adaptive purpose of acetylation in *P. putida*.** Acetylation is genetically and biochemically established in *P. fluorescens*/Achromobacter (PMID: 14507360; PMID: 37224964), but its specific contribution in *P. putida* (biofilm integrity vs. rhizosphere/desiccation fitness) has not been dissected.

4. **Cross-organism extrapolation risk.** The mechanistic backbone is assembled from *Rhodobacter* (BcsA–BcsB structure), *E. coli* (trans-envelope cryo-EM), *Gluconacetobacter* (reconstitution), and *Pseudomonas* (regulation/acetylation). Operon composition and product chemistry differ across these lineages, so uniform mechanistic claims should be made cautiously.
 5. **Crystallinity and higher-order assembly.** How lineage-specific factors (BcsD, cytoskeletal scaffolds) and modifications (acetylation, pEtN) tune fibre crystallinity and material properties remains incompletely understood ([PMID: 38688160](#); [PMID: 29926141](#)).
 6. **Regulatory integration.** How c-di-GMP pools are locally partitioned among competing effectors (cellulose vs. other EPS, motility appendages) to produce specific outputs is an active question in the broader c-di-GMP field.
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8. Key References

PMID	Contribution to this review
23222542	Crystal structure of <i>R. sphaeroides</i> BcsA–BcsB with translocating glucan; defines the two-subunit catalytic/translocation core (F001)
25942604	QM/MM mechanism of glucan elongation: SN2, ~68 kJ/mol barrier, ~8 s ⁻¹ (F001)
24704788	c-di-GMP activation: salt-bridge/gating-loop release; constitutive-activity mutants (F002)
26958837	In-crystallo enzymology: full biosynthetic cycle + "finger helix" translocation ratchet (F002)
31604608	BcsC structure: 16-strand β -barrel + 19 TPR OM export channel (F003)
39242554	Cryo-EM trans-envelope complex; BcsB semicircle tethers BcsC (F003)
30889223	<i>P. putida</i> PbcS repressed by FleQ/FleN, antagonized by c-di-GMP (F004)
21507177	<i>P. putida</i> cellulose: minor biofilm role, rhizosphere/desiccation fitness (F004)
14507360	WssFGHIJ acetylation genetics; acetylated cellulose required for WS biofilm (F005)
37224964	WssI is an O-acetyltransferase/acetylsterase on cello-oligomers (F005)
16151196	LPS–cellulose interactions govern biofilm strength (F005)
28131895	Wss-type acetylation operon distribution; HGT as single unit; <i>P. putida</i> carries it
25285473	Reconstitution in <i>E. coli</i> : activity requires co-expressed diguanylate cyclase
24139443	BcsA vs. plant CESA: conserved core, divergent assembly/regulation
29674724	<i>Gluconacetobacter</i> Type I operon (bcsABCD, BcsD); c-di-GMP as key activator
38688160	Review: synthase-dependent EPS secretion; crystallinity and modification variation
38645035	pEtN-cellulose modification/secretion (variant boundary)
28258142	Alginate polymerase / AlgFIJ acetylation (evolutionary template for WssGHI)
37500617	Gram-positive mixed-linkage β -glucan synthase (alternative route boundary)

Limitations of This Review

This synthesis draws its mechanistic backbone from **heterologous model organisms** — *Rhodobacter sphaeroides*, *E. coli*, and *Gluconacetobacter/Komagataeibacter* — and maps those insights onto the *P. putida* KT2440 module by orthology. The three *P. putida*-specific assignments in the working scope (bcsA, bcsB, PP_2638) are supported directly only at the level of regulation (FleQ/c-di-GMP) and gross physiological function (rhizosphere/desiccation), not at the level of protein structure or reconstituted activity in KT2440 itself. Claims about BcsB's c-di-GMP-binding role and PP_2638's export function should therefore be read as **well-motivated inferences**, not settled facts. No original experimental data were generated; the review is a literature-anchored synthesis of 28 papers across five iterations.

Proposed Follow-up Experiments / Actions

1. **Directly test PP_2638 function in KT2440.** Construct a clean Δ PP_2638 deletion and assay cellulose secretion (Calcofluor/Congo Red binding, cellulase-sensitive fibre EM), rhizosphere competition, and desiccation survival. Complement to confirm.
2. **Resolve BcsB's c-di-GMP role.** Purify *P. putida* BcsB periplasmic/cytoplasmic domains and test c-di-GMP binding (ITC/DRAcALA) to determine whether it is an independent sensor or primarily a structural organizer, discriminating between the annotation and the *E. coli*-derived model.
3. **Dissect acetylation phenotypes in *P. putida*.** Generate Δ wssFGHI (non-acetylating) mutants and quantify acetylation (NMR/MS), biofilm mechanics, LPS–cellulose interaction, and rhizosphere fitness to test whether acetylation matters as it does in *P. fluorescens*.
4. **Map the regulatory feed-forward.** Use controllable c-di-GMP levels (inducible DGC/PDE) with transcriptional reporters (PbcsA/PbcsD) and single-cell imaging to confirm that transcriptional derepression and catalytic activation are coincident in KT2440.
5. **Cryo-EM of the native *Pseudomonas* trans-envelope complex.** Determine whether the BcsB-semicircle/BcsC architecture seen in *E. coli* holds in the Wss-type system and where the acetylation machinery docks.
6. **Structure/function of PP_2638.** AlphaFold modeling followed by structural validation (Phenix/MolProbity) to confirm the predicted 16-strand β -barrel + TPR architecture and identify the aromatic-lined lumen.

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